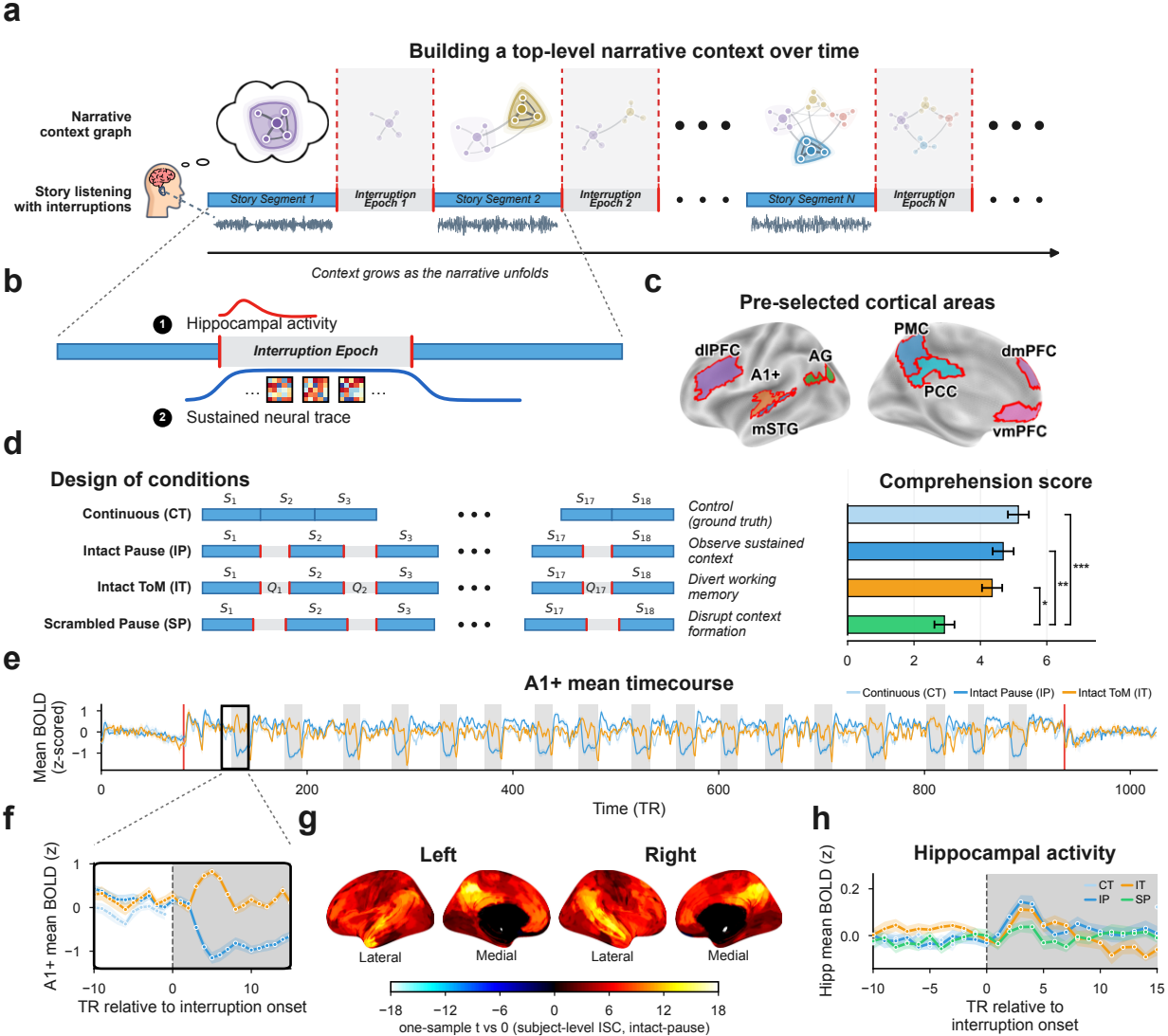

a
Building a top-level narrative context over time
Narrative context graph
Story listening with interruptions
Story Segment 1
Interruption Epoch 1
Story Segment 2
Interruption Epoch 2
Story Segment N
Interruption Epoch N
Context grows as the narrative unfolds
b
1 Hippocampal activity
Interruption Epoch
2 Sustained neural trace
c
Pre-selected cortical areas
dlPFC
A1+
AG
mSTG
PMC
dmPFC
PCC
vmPFC
d
Design of conditions
Continuous (CT)
Intact Pause (IP)
Intact ToM (IT)
Scrambled Pause (SP)
Control (ground truth)
Observe sustained context
Divert working memory
Disrupt context formation
Comprehension score
e
A1+ mean timecourse
Continuous (CT)
Intact Pause (IP)
Intact ToM (IT)
Mean BOLD (z-scored)
Time (TR)
f
A1+ mean BOLD (z)
TR relative to interruption onset
g
Left
Right
Lateral
Medial
one-sample t vs 0 (subject-level ISC, intact-pause)
h
Hippocampal activity
Hipp mean BOLD (z)
CT
IT
IP
SP
TR relative to interruption onset